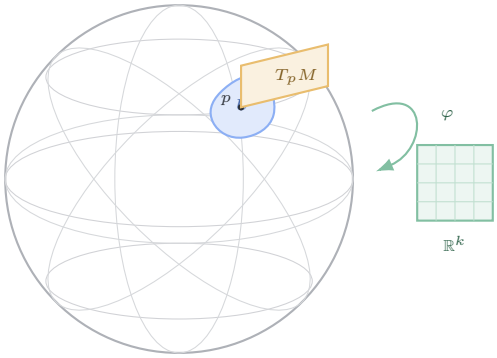
[tcb@cnt@defbox]Proposition [tcb@cnt@defbox]Lemma [tcb@cnt@defbox]Corollary

Lecture Notes

MAT257

Analysis on Manifolds



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How this notebook works

Notes are grouped **by day** (Day 1, Day 2, ...) rather than by topic, matching the order things were covered in lecture. Only three box types carry color — **Definition** (blue), **Theorem** (amber), and **Example** (green) — kept soft and light so they're easy on the eyes. Everything else (Lemma, Proposition, Corollary, Remark, Proof) is plain text with a bold label, no box.

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1 Day 1 (9.9) — Continuity and Limit (Tentative)

Definition 1.1: Continuity (Tentative)

A scalar function $f(x_1, \cdots, x_n)$ is continuous if it's continuous on function of each x_i. More generally, vector-valued function $f(x_1, \cdots, x_n)$ is continuous if all $f_j(x_1, \cdots, x_n)$ are continuous.

Example 1.2: Counterexample

Consider the function:

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2}, (x,y) \neq (0,0) \\ 0, (x,y) = (0,0) \end{cases}.$$

It is not continuous, although it is continuous for both x and y.

Definition 1.3: Limit (Tentative)

A scalar function $f : \mathbb{R}^n \to \mathbb{R}$ has its limit exist at $(x_1, \cdots, x_n)$ if for every direction vector $(a_1, \cdots, a_n)$,

$$\lim_{t \to 0} f(x_1 + ta_1, \cdots, x_n + ta_n)$$

exists and does not depend on $a = (a_1, \cdots, a_n)$.

Example 1.4: Counterexample

Consider the function:

$$f(x,y) = \begin{cases} \dfrac{y}{x^2}.x \neq 0 \\ 0, x = 0 \end{cases}.$$

We take the direction (a_1, a_2) such that $a_1 \neq 0$, and we plug this into f and we have:

$$f(ta_1, ta_2) = \frac{ta_2}{t^2 a_1^2} = \frac{a_2}{ta_1^2},$$

and when $t \to 0$, the expression goes to $\pm\infty$, and the limit does not exist.

Definition 1.5: Continuity (Tentative)

Let $x = (x_1, \cdots, x_n) \in \mathbb{R}^n$, then $f : \mathbb{R}^n \to \mathbb{R}$ is continuous at $x \in \mathbb{R}^n$ if:

$$\lim_{h \to 0} f(x + h) = f(x).$$

2 Day 2 (9.11) — Equivalent Norms and Continuity

Definition 2.1: Norm

A norm on a vector space is a function $\|\cdot\| : V \to \mathbb{R}_{\geq 0}$ such that:

1. $\|v\| = 0$ iff $v = 0$.
2. $\|tv\| = |t|\|v\|$ for all $t \in \mathbb{R}$ and $v \in V$.
3. Triangle inequality: $\|v + w\| \leq \|v\| + \|w\|$ for all $v, w \in V$.

Since $\mathbb{R}^n$ is a vector space, we can define the standard norm of $\mathbb{R}^n$, we let $x = (x_1, \cdots, x_n)$ so we have

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

Another important norm of $\mathbb{R}^n$ is the sup-norm:

$$\|x\|_\infty = \max\{|x_1|, \cdots, |x_n|\}.$$

Consider the norms:
If $\|x\| \leq R$, then x is contained in a closed ball with radius R. If $\|x\|_\infty \leq R$, then x is contained in a closed rectangle $[-R, R]^n$. An example of a norm on an infinite dimensional vector space $C([a, b])$ such that $[a, b] \subseteq \mathbb{R}$ is $\|f\|_\infty = \sup\{|f(x)| : x \in [a, b]\}$.

Theorem 2.2

Let $x \in \mathbb{R}^n$, then $\|x\|_\infty \leq \|x\| \leq \sqrt{n}\|x\|_\infty$.

Proof. We know that $\|x\|_\infty = \max\{|x_1|, \ldots, |x_n|\}$. For the lower bound, since each term is non-negative, we have:

$$\|x\|_\infty^2 = \max_i |x_i|^2 \leq \sum_{i=1}^n x_i^2 = \|x\|^2,$$

which implies $\|x\|_\infty \leq \|x\|$.

For the upper bound, summing over all n components gives:

$$\|x\|^2 = \sum_{i=1}^n x_i^2 \leq \sum_{i=1}^n \left(\max_j |x_j|\right)^2 = n\|x\|_\infty^2.$$

Taking the square root on both sides yields $\|x\| \leq \sqrt{n}\|x\|_\infty$. Thus, $\|x\|_\infty \leq \|x\| \leq \sqrt{n}\|x\|_\infty$. □

Definition 2.3: Equivalent Norms

Two norms $\|\cdot\|_1, \|\cdot\|_2$ on vector space V are equivalent if:

$$\exists c, C > 0 \text{ s.t. } c\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1.$$

Example 2.4

$\|\cdot\|_\infty$ and $\|\cdot\|$ are equivalent norms on $\mathbb{R}^n$ since we can choose $c = 1, C = \sqrt{n}$ to make

$$\|x\|_\infty \leq \|x\| \leq \sqrt{n}\|x\|_\infty.$$

Lemma 2.5. Any two norms on $\mathbb{R}^n$ are equivalent.

Definition 2.6: Continuity

A function $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is continuous at $x_0 \in A$ if:

$$\forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } \|x - x_0\| < \delta \Rightarrow \|f(x) - f(x_0)\| < \varepsilon.$$

f is continuous if it is continuous everywhere.

There, we can replace the Euclidean norm $\|\cdot\|$ by any equivalent norm. Also, the subset A is any subset of $\mathbb{R}^n$. Moreover, if $f : A \to \mathbb{R}^n$ is continuous and $A_1 \subseteq A$, then $f|_A$ is continuous.

Proposition 2.7 (Composition of Continuous Functions is Continuous)**.** Let $A \subseteq \mathbb{R}^n, B \subseteq \mathbb{R}^m$, and f, g be functions such that $f : A \to \mathbb{R}^m, g : B \to \mathbb{R}^k$ with $f(A) \subseteq B$. If f is continuous at x_0 and g is continuous at $f(x_0)$, then $g \circ f : A \to \mathbb{R}^k$ is continuous at x_0. In particular, if f, g are continuous, then $g \circ f$ is continuous.

Proof. Let $\varepsilon > 0$ be given. Since g is continuous at $f(x_0)$, then $\exists \delta_g > 0$ such that:

$$\|y - f(x_0)\| < \delta_g \Rightarrow \|g(y) - (g \circ f)(x)\| < \varepsilon.$$

Since f is continuous at x_0, then $\exists \delta_f > 0$ such that:

$$\|x - x_0\| < \delta_f \Rightarrow \|f(x) - f(x_0)\| < \varepsilon.$$

We take the radius corresponding to δ_g for f, so that:

$$\|x - x_0\| < \delta_f \Rightarrow \|f(x) - f(x_0)\| < \delta_g,$$

and this implies that:

$$\|x - x_0\| < \delta_f \Rightarrow \|(g \circ f)(x) - (g \circ f)(x_0)\| < \varepsilon,$$

as desired. □

Proposition 2.8. The set of continuous functions $f : A \to \mathbb{R}^m$ is a vector space.

Proof. As an exercise. □

Proposition 2.9. A function $f : A \to \mathbb{R}^m$ can be expressed as $f = (f_1, \cdots, f_n)$ for $i \in \{1, \cdots, n\}$, then f is continuous iff all of its component functions $f_i : A \to \mathbb{R}$ are continuous.

Proof. As an exercise. Hint: use the sup-norm $\|f(x) - f(x_0)\|_\infty$. □

3 Day 3 (9.14) — Topology on $\mathbb{R}^n$

Definition 3.1: Open Ball

The open ball around $x \in \mathbb{R}^n$ is defined as: $\mathrm{B}_\varepsilon(x) = \{y \in \mathbb{R}^n \mid \|x - y\| < \varepsilon\}$.

Definition 3.2: Open Set

A subset $U \subseteq \mathbb{R}^n$ is an open set if $\forall x \in U, \exists \varepsilon > 0$ s.t. $B_\varepsilon(x) \subseteq U$.

Definition 3.3: Open Neighborhood

An open neighborhood of $x \in \mathbb{R}^n$ is an open subset U with $x \in U$.

Definition 3.4: Open Ball

Let $x \in \mathbb{R}^n, \varepsilon > 0$, we call $B_\varepsilon(x) = \{y \in \mathbb{R}^n \mid \|x - y\| < \varepsilon\}$ the open ball around x.

Definition 3.5: Topological Continuity

A function $f : A \to \mathbb{R}^m$ be continuous at $x_0 \in A$ if $\forall \varepsilon > 0, \exists \delta > 0$ such that:

$$f(A \cap B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0)).$$

Proposition 3.6. 1. Every finite intersection of open sets is open.

2. Every union of open sets is open.

3. $\emptyset, \mathbb{R}^n$ are open.

Proof. 1. Suppose $U_1, \cdots, U_k \subseteq \mathbb{R}^n$ are open sets, let $x \in \bigcap_{i=1}^k U_i$, then there exists $\varepsilon_i > 0$ such that $B_{\varepsilon_i}(x) \subseteq U$. Let $\varepsilon = \min\{\varepsilon_1, \cdots, \varepsilon_k\}$, then $B_\varepsilon(x) \subseteq \bigcap_{i=1}^k U_i$.

2. Let $\{U_\alpha \mid \alpha \in I\}$ be a collection of open sets, where I is any $>$ set. Let $x \in \bigcup_{x \in I} U_\alpha$, and choose $\beta \in I$ with $x \in U_\beta$. Let $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq U_\beta \subseteq \bigcup_{\alpha \in I} U_\alpha$.

3. As an exercise.

□

Proposition 3.7. A function $f : A \to \mathbb{R}^m$ is continuous iff for all open sets $U \subseteq \mathbb{R}^m$, there exists open set $V \subseteq \mathbb{R}^n$ such that $f^{-1}(U) = A \cap V$. If A itself is open, $f : A \to \mathbb{R}^m$ is continuous iff for all open sets $U \subseteq \mathbb{R}^m$, $f^{-1}(U)$ is open.

Proof. ($\Rightarrow$) Suppose f is continuous, and $U \subseteq \mathbb{R}^m$ is open, then for all $x \in f^{-1}(U)$:

$$\exists \varepsilon_x > 0 \text{ s.t. } B_{\varepsilon_x}(f(x)) \subseteq U.$$

By continuity, we can find a $\delta_x > 0$, such that:

$$f(B_{\delta_x}(x) \cap A) \subseteq B_{\varepsilon_x}(f(x)) \subseteq U.$$

Let $V = \bigcup_{x \in f^{-1}(U)} B_{\delta_x}(x)$, then V is open and $A \cap V = \bigcup_{x \in f^{-1}(U)} B_{\delta_x}(x) \cap A \subseteq f^{-1}(U)$. Similarly, we also have $A \cap V \supseteq f^{-1}(U)$ since $x \in B_{\delta_x}(x)$. So we have:

$$A \cap V = f^{-1}(U).$$

($\Leftarrow$) Let $f : A \to \mathbb{R}^m$ be a function. Let $x_0 \in A$, let $\varepsilon > 0$ and let $U = B_\varepsilon(f(x_0))$. By assumption, there exists open sets $V \subseteq \mathbb{R}^n$ such that $f^{-1}(U) = A \cap V$. Since V is open, and $x_0 \in V$, we can choose $\delta > 0$ such that $B_\varepsilon(x_0) \subseteq V$, then:

$$f^{-1}(B_\varepsilon(f(x_0)) = A \cap V \supseteq A \cap B_\delta(x).$$

Hence $f(A \cap B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0))$, i.e. f is continuous at x_0. □

Definition 3.8: Closed Set

A subset $A \subseteq \mathbb{R}^n$ is closed if its complement $A^C = \mathbb{R}^n \backslash A$ is open.

Proposition 3.9. 1. Finite union of closed subsets are closed.

2. Arbitrary intersections of closed subsets are closed.

3. $\emptyset, \mathbb{R}^n$ are closed.

Proof. As an exercise. Hint: take complement. □

Definition 3.10: Clopen Set

A subset $A \subseteq \mathbb{R}^n$ is clopen if it is both closed and open.

Lemma 3.11. $\emptyset$ and $\mathbb{R}^n$ are the only clopen sets in $\mathbb{R}^n$.

Proof. Let $S \subseteq \mathbb{R}^n$ be a clopen set and assume $S \neq \emptyset$. Since S is clopen, it is closed, so that $S^C = \mathbb{R}^n \backslash S$ is open. Since S is also open, then $S \cup S^C = \mathbb{R}^n$ is open. Since $S \cap S^C = \emptyset$, and $\mathbb{R}^n$ is connected, by definition, there does not exists two disjoint nonempty open set such that their union equals to $\mathbb{R}^n$, so the only solution is $S = \mathbb{R}^n$ and $S^C = \emptyset$. □

4 Day 4 (9.16) — Interior, Closure, Connectedness and Compactness

Theorem 4.1: L

et $S \subseteq \mathbb{R}^n$ be a subset, then:

1. There exists an unique smallest closed subset containing S.
2. There exists an unique largest open subset contained in S.

These two subsets, are called the closure and the interior of S, denoted $\bar{S}$ and $\text{int}(S)$ respectively, with:

$$\text{int}(S) \subseteq S \subseteq \bar{S}.$$

Proof. 1. Defined $\bar{S} = \bigcap_{A \text{ is closed and } S \subseteq A} A$, so S is closed, and has the desired property.

2. Define $\text{int}(S) = \bigcup_{U \text{ is open and } U \subseteq S} U$, then U has the desired property.

□

Example 4.2

1. Let $S = \mathbb{Q}$, then $\text{int}(S) = \emptyset$ since the only open subset contained in S is $\emptyset$. Also, $\bar{S} = \mathbb{R}$ since every closed subset containing $\mathbb{Q}$ is $\mathbb{R}$.
2. Let $S = \{\frac{1}{n}, n \in \mathbb{N}\} \subseteq \mathbb{R}$, then $\text{int}(S) = \emptyset$ and $\bar{S} = S \cup \{0\}$.

Definition 4.3: Disconnected Set

A set $A \subseteq \mathbb{R}^n$ is disconnected if there are two open sets $U, V \subseteq \mathbb{R}^n$ such that:

$$U \cap V = \emptyset, A \cap U \neq \emptyset, A \cap V \neq \emptyset.$$

Definition 4.4: Connected Set

A set $A \subseteq \mathbb{R}^n$ is connected if it is not disconnected, that is, if there does not exists two open sets $U, V \subseteq \mathbb{R}^n$ such that:

$$U \cap V = \emptyset, A \cap U \neq \emptyset, A \cap V \neq \emptyset.$$

Example 4.5

1. Let $A_1 = \mathbb{R} \backslash \{0\}$, then A_1 is disconnected, since we can take:
$$U = \{x \mid x > 0\}, V = \{x \mid x < 0\},$$
such that $U \cap V = \emptyset$, $A_1 \cap U \neq \emptyset$ and $A_1 \cap V \neq \emptyset$.
2. Let $A_2 = \mathbb{R}^2 \backslash \{(0,0)\} \subseteq \mathbb{R}^2$, then A_2 is connected.
3. Let $A_3 = \mathbb{R}^2 \backslash \mathbb{S}^1 \subseteq \mathbb{R}^2$, where $\mathbb{S}^1 = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$. Then A_3 is disconnected, since we can take:
$$U = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\}, V = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 > 1\},$$
so that $U \cap V = \emptyset$, and $A_3 \cap U \neq \emptyset, A_3 \cap V \neq \emptyset$.

Definition 4.6: Path

A path is a subset $A \subseteq \mathbb{R}^n$, which is a continuous function $\gamma : [0,1] \to A$.

Definition 4.7: Path-Connected

A subset $A \subseteq \mathbb{R}^n$ is path-connected if for all $x, y \in A$, there exists a path $\gamma : [0,1] \to A$ such that $\gamma(0) = x, \gamma(1) = y$.

Lemma 4.8. 1. If a subset $A \subseteq \mathbb{R}^n$ is path-connected, then A is connected.

2. If a subset $U \subseteq \mathbb{R}^n$ is open, then U is path-connected iff U is connected.

Proof. 1. Suppose A is disconnected, we choose $U_1, U_2 \subseteq \mathbb{R}^n$, and consider $x \in A \cap U_1, y \in A \cap U_2$, pick a path $\gamma \in A$ from x to y, consider $\gamma^{-1}(U_1) \cup \gamma^{-1}(U_2) = [0,1]$, and we get contradiction, so A is connected.

2. As an exercise.

□

Example 4.9

There are some sets that are connected but not path-connected:

1. Let $X = \{(x, \sin(\frac{1}{x})) \mid 0 < x \leq 1\} \cup \{(0, y) \mid -1 \leq y \leq 1\}$, then X is connected but not path-connected.

Definition 4.10: Open Cover

An open cover of a subset $A \subseteq \mathbb{R}^n$ is a collection of open subsets $\{U_\alpha \mid \alpha \in I\}$ such that:

$$A \subseteq \bigcup_{\alpha \in I} U_\alpha.$$

Definition 4.11: Compact Set

A subset $C \subseteq \mathbb{R}^n$ is compact if for all open cover $\{U_\alpha \mid \alpha \in I\}$ of C, there exist finitely many $\alpha_1, \cdots, \alpha_r$ such that:

$$C \subseteq \bigcup_{i=1}^{r} U_{\alpha_i}.$$

5 Day 5 (9.18) — Heine-Borel Theorem

Definition 5.1: Bounded

A set $A \subseteq \mathbb{R}^n$ is bounded if there exists $R > 0$ such that $A \subseteq B_R(0)$.

Theorem 5.2: Heine-Borel Theorem

A set $C \subseteq \mathbb{R}^n$ is compact iff C is closed and bounded.

Proof. ($\Rightarrow$) Assume C is compact, consider the open cover by $\{B_R(0) \mid R > 0\}$. By compactness, there exists finitely many suffice. Let $\cup_{i=1}^n B_{R_i}(0) \supset C$, then we take $R = \max\{R_i\}$, then $C \subset B_R(0)$, thus C is bounded.
To show that C is closed, we have to show $\mathbb{R}^n \backslash C$ is open. Thus we have to show that for all $x \in \mathbb{R}^n \backslash C$, there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subset \mathbb{R}^n \backslash C$.
Let $x \in \mathbb{R}^n \backslash C$, then for every $y \in C$, let $r_= \frac{1}{3}\|x - y\|$, and cover C by $\{B_{r_y}(y) \mid y \in C\}$. Since C is compact, there exists a finite subcover $B_{r_{y_1}}, \cdots, B_{r_{y_n}}$ such that $\cup_{i=1}^n B_{r_{y_i}}(y_i) \supset C$, then let $\varepsilon = \min\{r_{y_1}, \cdots, r_{y_n}\}$, then $B_\varepsilon(x) \cap B_{r_{y_i}}(y_i) = \emptyset$, then $B_\varepsilon(x) \cap \cup_{i=1}^n B_{r_{y_i}}(y_i) = \emptyset$, thus $B_\varepsilon(x) \cap C = \emptyset$, so $B_\varepsilon(x) \subseteq \mathbb{R}^n \backslash C$.

($\Leftarrow$) Suppose C is closed and bounded, let $\{U_\alpha \mid \alpha \in I\}$ be an open cover, WTS $\exists U_{\alpha_1}, \cdots, U_{\alpha_n}$ such that $\cup_{i=1}^n U_{\alpha_i} \supset C$.
Suppose that there does not exist such finite subcover, since C is bounded, we choose a cube of side length R that contains it, i.e.

$$C \subseteq A = \Pi_{i=1}^n [a_i, b_i], b_i - a_i = R,$$

and we subdivide A into 2^n subcubes of the side length $\frac{1}{2}R$. Let $A_1 \subseteq A$ be a subcube such that $A_1 \cap C$ cannot be covered by finitely many open set U_α's. Thus, we iterate this process to create cubes $A \supseteq A_1 \supseteq A_2 \supseteq \cdots$, where A_k has side length $\frac{R}{2^k} \to 0$, and $A_k \cap C$ cannot be cover by finite many open set U_α's. Write $A_k = \prod_{i=1}^n [a_{k_i}, b_{k_i}]$, let $x_i = \lim_{k\to\infty} a_k = \lim_{k\to\infty} b_k \in \cap_k [a_{k_i}, b_{k_i}]$. Let $x = (x_1, \cdots, x_n) \in \bigcap_{k=1}^\infty A_k$. Thus, $x \in C$. Since C is closed, we can choose $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq \mathbb{R}^n \backslash C$, but $A_k \cap C \neq \emptyset$, which gives the contradiction, so $x \in C \cap \bigcap_{k=1}^\infty A_k$. Since $\{U_\alpha\}$ is a covering, we can choose α such that $x \in U_\alpha$. Since U_α is open, we can choose $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq U_\alpha$, and we can choose A_k with $A_k \subseteq B_\varepsilon(x) \subseteq U_\alpha$. So $A_k \cap C \subseteq U_\alpha$, thus this gives the contradiction

□

6 Day 6 (9.21) — Consequences of Compactness

Theorem 6.1: Images of Compact Sets Under Continuous Maps are Compact

Let $A \subseteq \mathbb{R}^n$ and suppose $f : A \to \mathbb{R}^m$ is continuous. Let $C \subseteq \mathbb{R}^n$ be compact with $C \subseteq A$, then $f(C)$ is compact.

Proof. Let $\{U_\alpha \mid \alpha \in I\}$ be an open cover of $f(C), since f is continuous, f^{-1}(U_\alpha) = A \cap V_\alpha$, where $V_\alpha \subseteq \mathbb{R}^n$ is open, and $\{V_\alpha \mid \alpha \in I\}$ is an open cover of C, then for $x \in C, f(x) \in U_\alpha$, $x \in f^{-1}(U_\alpha) = A \cap V_\alpha \subseteq V_\alpha$. Hence, there exists finite subcovers $V_1 \cdots, V_\alpha$ such that $f(C) \subseteq \bigcup_i V_i$ □

Corollary 6.2 (Extreme Value Theorem)**.** Suppose that $f : A \to \mathbb{R}$ is a continuous function, and let $C \subseteq \mathbb{R}^n$ be compact, with $C \subseteq A$, then $f|_C : C \to \mathbb{R}$ take on a maxima and minima on C.

Proof. Since $f(C) \subseteq \mathbb{R}$ is compact, by Heine-Borel Theorem, it is closed and bounded. Let $b = \sup(f(C)), a = \inf(f(C))$, since C is closed, and a, b are the minima and maxima of $f|_C$ on C. □

Theorem 6.3: ε-Neighborhood Theorem

Let $C \subseteq \mathbb{R}^n$ be a compact set, and let $U \subseteq \mathbb{R}^n$ be an open subset with $C \subseteq U$. Then, there exists $\varepsilon > 0$ such that the ε-neighborhood of C:

$$\{y \in \mathbb{R}^n : \|x - y\| < \varepsilon \text{ for some } x \in C\}$$

is centered in U.

Proof. For all $x \in C$, we choose $r_x > 0$ such that $B_{r_x} \subseteq U$, then $\{B_{\frac{r_x}{2}}(x) \mid x \in C\}$ covers C. By compactness, $B_{\frac{r_{x_1}}{2}}(x_1), \cdots, B_{\frac{r_{x_n}}{2}}(x_n)$ cover , so we let $\varepsilon = \min\{\frac{r_{x_1}}{2}, \cdots, \frac{r_{x_n}}{2}\}$. Now we suppose that there is an $y \in \mathbb{R}^n$, such that $x \in C$ with $\|x - y\| < \varepsilon$. Choose i such that $x \in B_{\frac{r_x}{2}}(x)$ and $\|x - x_i\| < \frac{r_x}{2}$, then $\|y - x_i\| \leq \|y - x\| + \|x - x_i\| < \varepsilon + \frac{r_{x_i}}{2} \leq r_{x_i}$, so $y \in B_{r_{x_i}}(x_i) \subseteq U$. □

Definition 6.4: Uniform Continuity

Let $A \subseteq \mathbb{R}^n$, then a function $f : A \to \mathbb{R}^m$ is uniformly continuous if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x, y \in A, \|x - y\| < \delta \Rightarrow \|f(x) - f(y)\| < \varepsilon.$$

Theorem 6.5

Let $C \subseteq \mathbb{R}^n$ be compact, and let $f : C \to \mathbb{R}^m$ be continuous, then f is uniformly continuous.

Proof. Let $\varepsilon > 0$ be given. For all $x \in C$, choose $r_x > 0$, such that for all $y \in C$, $\|x - y\| < r_x \Rightarrow \|f(x) = f(y)\| < \varepsilon$. And the sets $\{B_{\frac{r_x}{2}}(x) \mid x \in C\}$ covers C, and since C is compact, there exists finite subcover $B_{\frac{r_{x_1}}{2}}(x_1) \cdots, B_{\frac{r_{x_n}}{2}}(x_n)$, so let $\delta = \min\{\frac{r_{x_1}}{2}, \cdots, \frac{r_{x_n}}{2}\}$. Let $x, y \in C$, with $\|x - \|y < \delta$, then choose i with $x \in B_{\frac{r_{x_i}}{2}}(x_i)$, so $\|x - x_i\| < \frac{r_{x_i}}{2}$, and $\|y - x_i\| \leq \|y - x\| + \|x - x_i\| < \delta + \frac{r_{x_i}}{2} \leq r_{x_i}$, thus $\|f(x) - f(x_i)\| < \frac{\varepsilon}{2}$ and $\|f(y) - f(x_i)\| < \frac{\varepsilon}{2}$, giving that $\|f(x) - f(y)\| \leq \|f(x) - f(x_i)\| + \|f(y) - f(x_0\| = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. □

7 Day 7 (9.23) — Limit and Differentiability

Definition 7.1: Limit

Let $A \subseteq \mathbb{R}^n$ be a set and let $f : A \to \mathbb{R}^m$, let $c \in \bar{A}$, then we write:

$$\lim_{x \to c} f(x) = L,$$

if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x \in A, \|x - c\| < \delta \Rightarrow \|f(x) - L\| < \varepsilon$.
Note that we did not exclude $x = c$, and this convention is common in topology.

Example 7.2

Let:

$$f(x) = \begin{cases} 0, \text{if } x \neq 0, \\ 1, \text{if } x = 0 \end{cases},$$

then with our definition, $\lim_{x \to 0} f(x)$ does not exist, but with the other convention, we'd say $\lim_{x \to 0} f(x) = 0$.
With our definition:

$$\lim_{x \to 0, x \neq 0} f(x) = 0.$$

Proposition 7.3. Let $A \subseteq \mathbb{R}^n$, and a function $f : A \to \mathbb{R}^m$ is continuous function at $c \in A$ iff:

$$\lim_{x \to c} f(x) = f(c).$$

Recall that from MAT157 we defined that $f(x)$ is differentiable at $x = c$ if:

$$f'(c) = \lim_{x \to c, x \neq c} \frac{f(x) - f(c)}{x - c}.$$

In MAT257 we generalized this definition of differentiability into $\mathbb{R}^n$. Our idea is to pick an arbitrary point $a \in \mathbb{R}^n$ and consider a direction of a.

Definition 7.4: Directional Derivative

Let $U \subseteq \mathbb{R}^n$ be an open set and let $f : U \to \mathbb{R}^m$ be a function, then the directional derivative of f at $x = c$ with the directional vector $v \in \mathbb{R}^n$ is defined as:

$$D_v f(c) = \lim_{h \to 0, h \neq 0} \frac{f(c + hv) - f(c)}{h}.$$

Definition 7.5: Partial Derivative

Let e_i be the i-th standard basis vector, the i-th partial derivative of f, denoted by $\frac{\partial f}{\partial x_i}$, at $x = c$, is defined as:

$$\frac{\partial f}{\partial x_i}(c) = D_{e_i} f(c)$$

Equivalently:

$$\frac{\partial f}{\partial x_i}(c) = \lim_{h\to 0, h\neq 0} \frac{f(c+he_i)-f(c)}{h}.$$

Example 7.6

Let $f(x,y,z) = x\sin y + xyz^2$, then:

$$\frac{\partial f}{\partial x}(x,y,z) = \sin y + z^2 y,\ \frac{\partial f}{\partial y}(x,y,z) = x\cos y + z^2 x,\ \frac{\partial f}{\partial z}(x,y,z) = 2xyz$$

What does it mean for f to be differentiable at c? We know that in MAT157, f is differentiable at c then f is continuous at c, does it still hold in $\mathbb{R}^n$? We have some examples:

Example 7.7

f differentiable $\Rightarrow$ all partial derivatives exist, but $\Leftarrow$ fails. Let:

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, (x,y)\neq(0,0) \\ 0, (x,y) = (0,0) \end{cases}$$

, then all $\frac{\partial f}{\partial x_i}$ exist, but f is not continuous.

Example 7.8

f differentiable $\Rightarrow$ all directional derivatives exist, but $\Leftarrow$ fails. Let:

$$f(x,y) = \begin{cases} \frac{yx^2}{x^4+y^2}, (x,y)\neq(0,0) \\ 0, (x,y) = (0,0) \end{cases}$$

Let $v = (v_1, v_2)$, then $D_v f(0) = \begin{cases} \frac{v_1^2}{v_2}, v_2 \neq 0 \\ 0, v_2 = 0 \end{cases}$, all directional derivatives exist but f is not continuous.

And finally we define the differentiabililty of f at $x = c$.

Definition 7.9: Differentiable

Let $U \subseteq \mathbb{R}^n$ be an open set. Let $f: U \to \mathbb{R}^m$ be a function, then f is differentiable at $x = c$ if there exists a linear transformation $M: \mathbb{R}^n \to \mathbb{R}^m$ such that:

$$\lim_{h\to 0, h\neq 0} \frac{f(c+h) - f(c) - M(h)}{\|h\|} = 0.$$

8 Day 8 (9.25) — Differentiation

Definition 8.1: Differentiable

Let $U \subseteq \mathbb{R}^n$ be an open set, and let $f : U \to \mathbb{R}^m$ be a function, then f is differentiable at $c \in U$ if there exists a linear map $M : \mathbb{R}^n \to \mathbb{R}^m$ such that:

$$f(x) = f(c) + M(x - c) + r(x),$$

where $\lim_{x \to c, x \neq c} \dfrac{\|r(x)\|}{\|x - c\|} = 0.$

Lemma 8.2. The map $M : \mathbb{R}^n \to \mathbb{R}^m$ in the definition above (if it exists) is unique.

Proof. Suppose $f(x) = f(c) + M_1(x - c) + r_1(x) = f(c) + M_2(x - c) + r_2(x)$, where $r_1(x), r_2(x)$ satisfy $\lim_{x \to c} \dfrac{\|r_i(x)\|}{\|x - c\|} = 0$. Then:

$$(M_1 - M_2)(x - c) + r_1(x) - r_2(x) = 0$$

Hence:

$$\|(M_1 - M_2)(x - c)\| = \|r_1(x) - r_2(x)\|$$

So we divide by $\|x - c\|$, and we obtain:

$$\frac{\|(M_1 - M_2)(x - c)\|}{\|x - c\|} = \frac{\|r_1(x) - r_2(x)\|}{\|x - c\|} \leq \frac{\|r_1(x)\|}{\|x - c\|} + \frac{\|r_2(x)\|}{\|x - c\|}$$

Thus:

$$\lim_{x \to c} \frac{\|(M_1 - M_2)(x - c)\|}{\|x - c\|} = 0$$

Fix $a \neq 0$. Since U is open, $c + ta \in U$ for $|t|$ small, and $c + ta \neq c$ for $t \neq 0$. Then

$$\frac{\|(M_1 - M_2)(ta)\|}{\|ta\|} = \frac{|t| \, \|(M_1 - M_2)a\|}{|t| \, \|a\|} = \frac{\|(M_1 - M_2)a\|}{\|a\|},$$

which is independent of t. Its limit as $t \to 0$ is 0, so $(M_1 - M_2)a = 0$. □

Definition 8.3: Derivative

In this case, the linear map M is called the derivative of f at c, denoted $Df(c)$.

Example 8.4

If $f(x) = Mx + b$, where $M : \mathbb{R}^n \to \mathbb{R}^m$ is linear, and $b \in \mathbb{R}^m$, then f is differentiable everywhere, with $Df(c) = M$.

Proposition 8.5. Let $U \subseteq \mathbb{R}^n$ be an open set, and let $f : U \to \mathbb{R}^m$ be a function. If f is differentiable at $c \in U$, then it is continuous at c.

Proof. We know $f(x) = f(c) + Df(c)(x-c) + r(x)$, and $\lim_{x\to c, x\neq c} \dfrac{\|r(x)\|}{\|x-c\|} = 0$, then:

$$\|r(x)\| = \frac{\|r(x)\|}{\|x-c\|} \cdot \|x-c\| \to 0,$$

thus $\lim_{x\to c} r(x) = 0$. Also, since $Df(c)$ is linear (hence continuous),

$$\lim_{x\to c} Df(c)(x-c) = 0,$$

hence there is $\lim_{x\to c} f(x) = f(c)$, hence f is continuous at c. □

Proposition 8.6. Let $U \subseteq \mathbb{R}^n$ be an open set, and let $f : U \to \mathbb{R}^m$ be a function. If f is differentiable at $c \in U$, then all directional derivatives $D_v f(c)$ for vector $v \in \mathbb{R}^n$ exist, and:

$$D_v f(c) = Df(c) \cdot v$$

Proof. For $v = 0$, both sides equal to 0, so we assume $v \neq 0$, then let $x = c + tv$, we have:

$$\frac{f(c+tv) - f(c)}{t} = Df(c) \cdot v + \frac{r(c+tv)}{t},$$

but $\lim_{t\to 0} \frac{\|r(c+tv)\|}{\|t\|} = \|v\| \lim_{t\to 0} \frac{\|r(c+tv)\|}{\|tv\|} = 0$. Hence:

$$\lim_{t\to 0} \frac{f(c+tv) - f(c)}{t} = D_v f(c) = Df(c) \cdot v.$$

□

Taking $v = e_j$ in the proposition above, we get $Df(c)e_j = D_{e_j} f(c) = \frac{\partial f}{\partial x_j}(c) = \left(\frac{\partial f_1}{\partial x_j}(c), \ldots, \frac{\partial f_m}{\partial x_j}(c)\right)^T$. Since the j-th column of the matrix of the linear map $Df(c)$ is $Df(c)e_j$, the matrix of $Df(c)$ is:

$$Df(c) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(c) & \cdots & \frac{\partial f_1}{\partial x_n}(c) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(c) & \cdots & \frac{\partial f_m}{\partial x_n}(c) \end{pmatrix}$$

And it is called the Jacobian matrix of f at c.

9 Day 9 (9.28) — Curves, Gradients, and Level Sets

Example 9.1: Special Cases

1. When $n = 1$, a continuous map $\gamma : (a, b) \to \mathbb{R}^m$ is called a curve in $\mathbb{R}^m$. If γ is differentiable at $t_0 \in (a, b)$, then we call:

$$\gamma'(t_0) = D\gamma(t_0) = \begin{pmatrix} \gamma_1'(t_0) \\ \vdots \\ \gamma_m'(t_0) \end{pmatrix}$$

the velocity vector of γ at t_0, and $\|\gamma'(t_0)\|$ is called the speed of γ at t_0. Notice that if γ is differentiable everywhere, then $t \to \gamma'(t)$ is also a curve, then the second derivative, if it exists, $\gamma''(t)$ is called the acceleration. For curves with constant speed, they have zero acceleration.

2. When $m = 1$, the function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is called a scalar function. Suppose f is differentiable at c, then $Df(c)$ is a vector, where:

$$Df(c) = \left(\frac{\partial f}{\partial x_1}(c), \cdots, \frac{\partial f}{\partial x_n}(c) \right),$$

and this is called the gradient of f at c, denoted $\nabla f(c)$. The only difference between $Df(c)$ and $\nabla f(c)$ is that $Df(c)$ is the row vector and $\nabla f(c)$ is the column vector, i,e,

$$\nabla f(c) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(c) \\ \vdots \\ \frac{\partial f}{\partial x_n}(c) \end{pmatrix}.$$

Definition 9.2: Critical Point

A point $c \in U$ is a critical point of $f : U \in \mathbb{R}^n \to \mathbb{R}^m$ if $\nabla f(c) = 0$.

Proposition 9.3. Suppose f has a local maximum at $c \in U$, then:

$$\exists \varepsilon > 0 \text{ s.t. } \forall x \in B_\varepsilon(c), f(x) \leq f(c).$$

If f is differentiable at c, then $\nabla f(c) = 0$.

Proof. For any $c \in \mathbb{R}^n$, we let $g(t) = f(c + ta)$. For $|t|$ that is sufficiently small, we have:

$$G(t) \leq g(0) = f(c) \Rightarrow g'(0) = 0,$$

which gives:

$$D_a f(c) = \lim_{t \to 0} \frac{f(c + ta) - f(c)}{t} = 0 \Rightarrow \nabla f(c) = 0$$

□

Example 9.4

1. Let $f(x,y) = x^2 + y^2$, then the level sets $f^{-1}(c)$ are circles, and:
$$\nabla f(x,y) = \begin{pmatrix} 2x \\ 2y \end{pmatrix}.$$
2. Let $g(x,y) = x^2 - y^2$, then the level sets of g are the hyperbolas, and:
$$\nabla g(x,y) = \begin{pmatrix} 2x \\ -2y \end{pmatrix}.$$
3. Let $h(x,y) = y - x^2$, then the level sets of h are parabolas, and:
$$\nabla h(x,y) = \begin{pmatrix} -2x \\ 1 \end{pmatrix}.$$

Lemma 9.5. $\nabla f(x_0)$ is always orthogonal to the level set through x_0. More precisely, every differentiable curve $\gamma : \mathbb{R} \to \mathbb{R}^n$ with $\gamma(t_0) = x_0$ and $f(\gamma(t))$ is a constant, we have:
$$\langle \gamma'(t_0), \nabla f(\gamma(t_0)) \rangle = 0.$$

Proof. We take the derivative of $f(\gamma(t))$, and we obtain:
$$0 = \frac{d}{dt} f(\gamma(t)),$$
now by the Chain Rule:
$$0 = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\gamma(t)) \frac{d\gamma_i}{dt} = \langle \nabla f(\gamma(t)), \gamma'(t) \rangle.$$
□

10 Day 10 (9.30) — Continuous Differentiability

Lemma 10.1 (Necessary Condition for Differentiability)**.** Let $U \subseteq \mathbb{R}^n$ be an open set and let $f : U \to \mathbb{R}^m$ be a function. A necessary condition for the differentiability of f at $c \in U$ is that all $\frac{\partial f_i}{\partial x_j}(c)$ exist, but this condition is not sufficient.

Theorem 10.2: S

uppose $U \subseteq \mathbb{R}^n$ is an open set, and $f : U \to \mathbb{R}^m$ is a function such that all $\frac{\partial f_i}{\partial x_j}(x)$ exist for all $x \in U$. If they are all continuous on U, then f is differentiable on U.

Proof. Let $c \in U$ be given, and let M be the Jacobian matrix

$$M = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(c) & \cdots & \frac{\partial f_1}{\partial x_n}(c) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(c) & \cdots & \frac{\partial f_m}{\partial x_n}(c) \end{pmatrix}.$$

Define $r(x)$ by

$$f(x) = f(c) + M(x - c) + r(x);$$

we have to prove that $\lim_{x \to c} \dfrac{\|r(x)\|}{\|x - c\|} = 0$.

Choose $\delta > 0$ such that $B_\delta(c) \subseteq U$. We consider $x \in B_\delta(c)$; then the rectangular box with opposite corners c and x is contained in $B_\delta(c) \subseteq U$. Explicitly, we write $h = x - c = \sum_{i=1}^n h_i e_i$, where $e_i \in \mathbb{R}^n$ is the i-th standard basis vector. Set

$$c^{(0)} = c, \qquad c^{(j)} = c + \sum_{i=1}^{j} h_i e_i \quad (j = 1, \ldots, n),$$

so that $c^{(n)} = x$ and

$$f_i(x) = f_i(c) + \sum_{j=1}^{n} \big(f_i(c^{(j)}) - f_i(c^{(j-1)}) \big).$$

(Note: $c^{(j)}$ and $c^{(j-1)}$ differ only in the j-th coordinate.)

We'll use the Mean Value Theorem: if $g : [a, b] \to \mathbb{R}$ is continuous, and differentiable on (a, b), then there exists $\xi \in (a, b)$ such that

$$g(b) - g(a) = g'(\xi)(b - a).$$

Applying this to $f_i(c^{(j)}) - f_i(c^{(j-1)})$, there exists ξ_{ij} on the segment from $c^{(j-1)}$ to $c^{(j)}$ such that

$$f_i(c^{(j)}) - f_i(c^{(j-1)}) = \frac{\partial f_i}{\partial x_j}(\xi_{ij})\, h_j.$$

Thus

$$f_i(x) - f_i(c) = \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}(\xi_{ij})\, h_j.$$

On the other hand,

$$\big(M(x-c)\big)_i = \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}(c)\, h_j.$$

Hence

$$r_i(x) = \sum_{j=1}^{n} \left(\frac{\partial f_i}{\partial x_j}(\xi_{ij}) - \frac{\partial f_i}{\partial x_j}(c) \right) h_j \;\Rightarrow\; |r_i(x)| \le \sum_{j=1}^{n} \left| \frac{\partial f_i}{\partial x_j}(\xi_{ij}) - \frac{\partial f_i}{\partial x_j}(c) \right| |h_j|.$$

Let

$$a_{ij}(x) = \sup_{\|y-c\| \le \|x-c\|} \left| \frac{\partial f_i}{\partial x_j}(y) - \frac{\partial f_i}{\partial x_j}(c) \right|.$$

Since $\|\xi_{ij} - c\| \le \|x - c\|$, the summand above is at most $a_{ij}(x)|h_j|$. By continuity of the partial derivatives,

$$\lim_{y \to c} \left(\frac{\partial f_i}{\partial x_j}(y) - \frac{\partial f_i}{\partial x_j}(c) \right) = 0 \;\Rightarrow\; \lim_{x \to c} a_{ij}(x) = 0.$$

Let $a(x) = \max_{i,j} a_{ij}(x)$; then we also have $\lim_{x \to c} a(x) = 0$. Moreover,

$$|r_i(x)| \le a(x) \sum_{j=1}^{n} |h_j| = a(x) \sum_{j=1}^{n} |x_j - c_j| \le \sqrt{n}\, a(x) \|x - c\|.$$

Thus

$$\frac{|r_i(x)|}{\|x-c\|} \le \sqrt{n}\, a(x) \;\Rightarrow\; \lim_{x \to c} \frac{|r_i(x)|}{\|x-c\|} = 0 \text{ for each } i \;\Rightarrow\; \lim_{x \to c} \frac{\|r(x)\|}{\|x-c\|} = 0.$$

□

Some notations:

- $C^0(U, \mathbb{R}^m)$ is the space of continuous functions $f : U \to \mathbb{R}^m$.
- $C^1(U, \mathbb{R}^m)$ is the space of continuously differentiable functions $f : U \to \mathbb{R}^m$.
- $C^r(U, \mathbb{R}^m)$ is the space of r-times continuously differentiable functions $f : U \to \mathbb{R}^m$.
- $C^\infty(U, \mathbb{R}^m)$ is the space of smooth (∞-times differentiable) functions $f : U \to \mathbb{R}^m$.

11 Day 11 (10.2) — Higher Order Partial Derivatives

Theorem 11.1: Equality of Mixed Partials

Let $U \subseteq \mathbb{R}^n$ be open and let $f \in C^2(U, \mathbb{R}^m)$, then for $i, j \in \{1, \cdots, n\}$:

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

Proof. □

12 Day 12 (10.5) — The Chain Rule

Theorem 12.1: The Chain Rule

Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open, and let:

$$f : U \to \mathbb{R}^m,\ g : V \to \mathbb{R}^k,\ \text{s.t. } f(U) \subseteq V.$$

If f is differentiable at c, and g is differentiable at $f(c)$, then $g \circ f : U \to \mathbb{R}^k$ is differentiable at c, and we have:

$$D(g \circ f)(c) = Dg(f(c))Df(c).$$

Proof. We write:

$$f(x) = f(c) + M(x-c) + r_f(x), \lim_{x\to c, x\neq c} \frac{r_f(x)}{\|x-c\|} = 0,$$

and that:

$$g(y) = g(f(c) + N(y - f(c)) + r_g(y), \lim_{y\to f(c), y\neq f(c)} \frac{r_g(y)}{\|y - f(c)\|} = 0,$$

and that $M = Df(c), N = Dg(f(c))$.
And we have that:

$$\begin{aligned} g(f(x)) &= g(f(c)) + N(f(x) - f(c)) + r_g(f(x)) \\ &= g(f(c)) + N(M(x-c) + r_f(x)) + r_g(f(x)) \\ &= g(f(c)) + NM(x-c) + (Nr_f(x) + r_g(f(x))) \\ &= g(f(c)) + NM(x-c) + r_{g\circ f}(x) \quad (*) \end{aligned}$$

where $\frac{r_{g\circ f}(x)}{\|x-c\|} = \frac{Nr_f(x)}{\|x-c\|} = \frac{r_g(f(x))}{\|x-c\|}$.
Let:

$$\sigma_f(x) = \begin{cases} \frac{r_f(x)}{\|x-c\|}, x \neq c \\ 0, x = c \end{cases}, \sigma_g(x) = \begin{cases} \frac{r_g(x)}{\|x-c\|}, x \neq c \\ 0, x = c \end{cases}$$

Then we have:

$$\frac{r_{g\circ f}(x)}{\|x-c\|} = Nr_f(x) + \sigma_g(f(x))\frac{\|f(x) - f(c)\|}{\|x-c\|}.$$

Since f is continuous at c, we have $\lim_{x\to c} \sigma_f(x) = \sigma_f(c) = 0$, and $\lim_{x\to c} \sigma_g(f(x)) = \sigma_g(f(c)) = 0$, thus:

$$\frac{\|f(x) - f(c)\|}{\|x-c\|} = \frac{\|M(x-c) + r_f(x)\|}{\|x-c\|} \leq \left\| M\frac{x-c}{\|x-c\|} \right\| + \frac{\|r_f(x)\|}{\|x-c\|}$$

But $\left\| M\frac{x-c}{\|x-c\|} \right\| \leq \max\{\|Ma\| : \|a\| = 1\} = c$, then we know that $\lim_{x\to c} \frac{r_{g\circ f}(x)}{\|x-c\|} = 0$. Therefore, by $(*)$ we have that $D(g \circ f)(c) = NM$. □

Proposition 12.2. Let $U \subseteq \mathbb{R}^n$ be open, let $f \in C^1(U)$ and $c \in U$, then $\nabla f(c) = (Df(c))^T$.
(*Observation: $\nabla f(c)$ is orthogonal to the level set of f)
More precisely, given any $\gamma : (a, b) \to \mathbb{R}^n$, $\gamma \in C^1(a, b)$, with $f(\gamma(t))$ is a constant, then for all $t \in (a, b)$, $\langle \nabla f(\gamma(t)), \gamma'(t) \rangle = 0$.

Proof. Take derivative of $f(\gamma(t))$ as a constant, then:

$$Df(\gamma(t))D\gamma(t) = D_{f(\gamma(t))}(t) = 0.$$

Therefore we have $\langle \nabla f(\gamma(t)), \gamma'(t) \rangle = 0$. □

Proposition 12.3 (Leibniz Notation)**.** Let $U \subseteq \mathbb{R}^n$ be open and given $f : U \to \mathbb{R}^n$, let $y = f(x)$, then we write $Df(x)$ as a matrix with entries $\dfrac{\partial x_i}{\partial y_j}$. Given another function $z = g(y)$, then $z = g(f(x))$, and by the Chain Rule, we have:

$$\frac{\partial z_i}{\partial x_k} = \sum_{j=1}^{m} \frac{\partial z_i}{\partial y_j} \frac{\partial y_j}{\partial x_k}$$

Example 12.4

Consider $\frac{d}{dt} \int_0^t e^{-ts^2} ds$, we write the funtion:

$$g(x, y) = \int_0^x e^{-ys^2} ds,$$

and we have to show $\frac{d}{dt} g(t, t)$, we have:

$$\frac{d}{dt} g(t, t) = \frac{\partial g}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial g}{\partial y} \frac{\partial y}{\partial t} = e^{-yx^2} - \int_0^x s^2 e^{-yx^2} ds$$

13 Day 13 (10.7) — The Chain Rule and Matrix Transformations

We know the Chain Rule:

$$D(g \circ f)(c) = Dg(f(c))Df(c)$$

And the Leibniz notation: when $z = g(y), y = f(x)$, we have:

$$\frac{\partial z_i}{\partial x_k} = \sum_{j=1}^{n} \frac{\partial z_i}{\partial y_j} \frac{\partial y_j}{\partial x_k}$$

And there are some examples:

Example 13.1

Let $f : \mathbb{R}^2 \to \mathbb{R}^3$ being represented by:

$$f(x_1, x_2) = (e^{x_1+x_2}, x_1^2 + x^2 + 1, \sin(x_1) + 2x_1)$$

And $g : \mathbb{R}^3 \to \mathbb{R}^2$ being represented by:

$$g(y_1, y_2, y_3) = (y_1 + y_2 y_3 + \sin(y_3), y_1^2 + y_2 + y_3^2 + 1)$$

And by the Chain Rule, we know that:

$$D(g \circ f)(c) = Dg(f(c))Df(c),$$

consider the point $c = \mathbf{0}$, we then have:

$$Df(c) = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y2}{\partial x_2} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 3 & 0 \end{pmatrix}.$$

And $f(\mathbf{0}) = (1, 1, 0)$, so that:

$$Dg(c) = \begin{pmatrix} \frac{\partial z_1}{\partial y_1} & \frac{\partial z_1}{\partial y_2} & \frac{\partial z_1}{\partial y_3} \\ \frac{\partial z_2}{\partial y_1} & \frac{\partial z_2}{\partial y_2} & \frac{\partial z_2}{\partial y_3} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \end{pmatrix}.$$

Therefore, we have:

$$D(g \circ f)(c) = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} 7 & 1 \\ 2 & 3 \end{pmatrix}.$$

Example 13.2: Applications On the Product Rule

Consider the value of $\frac{d}{dt}(fg)$, we let $x = f(t), y = g(t)$, then:

$$\frac{d}{dt}(fg) = \frac{d}{dt}(xy) = \frac{\partial}{\partial x}(xy)\frac{dx}{dt} + \frac{\partial}{\partial y}(xy)\frac{dy}{dt} = g(t)\frac{df}{dt} + f(t)\frac{dg}{dt}.$$

Proposition 13.3. Let $f : U \to \mathbb{R}^m$ be differentiable at $c \in U$ and let $a \in \mathbb{R}^n$. If $\gamma : (-\varepsilon, \varepsilon) \to U$ is differentiable, with $\gamma(0) = c, \gamma'(0) = a$, then:

$$D_a f(c) = \frac{d}{dt}_{t=0} f(\gamma(t)).$$

Proof. $\frac{d}{dt}_{t=0} f(\gamma(t)) = Df(\gamma(0))D\gamma(0) = Df(c) \cdot a \cdot \gamma'(0) = D_a f(c)$. □

Example 13.4: Application to Matrices

We know that $\mathrm{Mat}_{\mathbb{R}}(m \times n) \cong \mathbb{R}^{m \times n}$, and we consider the following functions:

1. $\det : \mathrm{Mat}_{\mathbb{R}}(n \times n) \to \mathbb{R}$
2. $\mathrm{Mult} : \mathrm{Mat}_{\mathbb{R}}(n \times n) \times \mathrm{Mat}_{\mathbb{R}}(n \times n) \to \mathrm{Mat}_{\mathbb{R}}(n \times n)$
3. $\mathrm{Inv} : \{A : \det(A) \neq 0\} \to \mathrm{Mat}_{\mathbb{R}}(n \times n)$
4. $\exp : \mathrm{Mat}_{\mathbb{R}}(n \times n) \to \mathrm{Mat}_{\mathbb{R}}(n \times n)$

We consider the derivatives:

1. WTS: $D\,\mathrm{Mult}(A_1, A_2)$.
   We first consider $A_1 = A_2 = I$, then:

$$\begin{aligned} D\,\mathrm{Mult}(I, I)(B_1, B_2) &= \frac{d}{dt}_{t=0} \mathrm{Mult}(I + tB_1, I + tB_2) \\ &= \frac{d}{dt}_{t=0} (I + t(B_1 + B_2) + t^2 B_1 B_2) \\ &= B_1 + B_2 \end{aligned}$$

   And in general:

$$\begin{aligned} D\,\mathrm{Mult}(A_1, A_2)(B_1, B_2) &= \frac{d}{dt}_{t=0} \mathrm{Mult}(A_1 + tB_1, A_2 + tB_2) \\ &= \frac{d}{dt}_{t=0} (A_1 A_2 + t(B_1 A_2 + A_1 B_2) + t^2 B_1 B_2 \\ &= B_1 A_2 + A_1 B_2 \end{aligned}$$

   Then $D\,\mathrm{Mult}(A_1, A_2)$ is the linear map $\mathrm{Mat}(n \times n) \times \mathrm{Mat}(n \times n) \to \mathrm{Mat}(n \times n), (B_1, B_2) \to B_1 A_2 + A_1 B_2$.

2. WTS: $D\,\mathrm{Inv}(A)$. We consider that $\mathrm{Inv}(A) + A^{-1}$ defined on $U = \{A : \det(A) \neq 0\}$. Then, we have:

$$\begin{aligned} D\,\mathrm{Inv}(A)(B) &= \frac{d}{dt}_{t=0} \mathrm{Inv}(A + tB) \\ &= \frac{d}{dt}_{t=0} (A + tB)^{-1} \\ &= -A^{-1} B A^{-1} \end{aligned}$$

14 Day 14 (10.9) — Polar Coordinates, Differentiable Curves and Matrix Differentials

Example 14.1: Polar Coordinate

We let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ and $(r, \theta) \to (r\cos\theta, r\sin\theta)$, so that every $(x, y) \in \mathbb{R}^2$ has preimage (r, θ) with $r \geq 0$, and θ determined up to the mutiples of 2π. Given $u = f(x, y)$, then we have:

$$\begin{aligned}\frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} \\ &= \cos\theta\frac{\partial u}{\partial x} + \sin\theta\frac{\partial u}{\partial y} \\ &= \frac{x}{r}\frac{\partial u}{\partial x} + \frac{y}{r}\frac{\partial u}{\partial y}\end{aligned}$$

And similarly:

$$\begin{aligned}\frac{\partial u}{\partial \theta} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta} \\ &= -r\sin\theta\frac{\partial u}{\partial x} + r\cos\theta\frac{\partial u}{\partial y} \\ &= -y\frac{\partial u}{\partial x} + x\frac{\partial u}{\partial y}\end{aligned}$$

Then we have $e_r = (\cos\theta, \sin\theta) = \frac{1}{r}(x, y)$ and $e_\theta = (-\sin\theta, \cos\theta) = \frac{1}{r}(-y, x)$, thus:

$$\frac{\partial u}{\partial r} = \langle \nabla u, e_r\rangle, \frac{\partial u}{\partial \theta}\langle \nabla u, e_\theta\rangle.$$

Hence:

$$\nabla u = \frac{\partial u}{\partial r}e_r + \frac{1}{r}\frac{\partial u}{\partial \theta}e_\theta$$

Example 14.2: Velocity Vectors of Differentiable Curves

Let $\gamma : \mathbb{R} \to \mathbb{R}^2$ be differentiable, and $\gamma(t) = (x(t), y(t))$, then $\gamma'(t) = (\frac{dx}{dt}, \frac{dy}{dt})$, then we express it in polar coordinates by letting $\gamma(t) = \phi(r(t), \theta(t))$, then:

$$\frac{dx}{dt} = \frac{\partial x}{\partial r}\frac{dr}{dt} + \frac{\partial x}{\partial \theta}\frac{d\theta}{dt} = \cos\theta\frac{dr}{dt} - r\sin\theta\frac{d\theta}{dt}$$

Similarly:

$$\frac{dy}{dt} = \sin\theta\frac{dr}{dt} + r\cos\theta\frac{d\theta}{dt}$$

So $\gamma'(t) = (\frac{dx}{dt}, \frac{dy}{dt}) = \frac{dr}{dt}e_r + r\frac{d\theta}{dt}e_\theta$, and $\|\gamma'(t)\| = \sqrt{(\frac{dr}{dt})^2 + r^2(\frac{dy}{dt})^2}$

Recall that $\mathrm{Mat}_{\mathbb{R}}(n \times n) = \mathbb{R}^{n^2}$, we have:

$$U = \{A \mid \det(A) \neq 9\} \subseteq \mathrm{Mat}_{\mathbb{R}}(n \times n),$$

and that $\mathrm{Inv} : U \to U \subseteq \mathrm{Mat}_{\mathbb{R}}(n \times n)$, with $D\,\mathrm{Inv}(A)(B) = -A^{-1}BA^{-1}$.

Example 14.3

Suppose $\gamma : \mathbb{R} \to U$, and $\gamma(t) = A(t)$, sd sts $\frac{d}{dt}A(t)^{-1}$ in terms of $\frac{dA}{dt}$. We have:

$$0 = \frac{d}{dt}(I) = \frac{d}{dt}(A(t)A(t)^{-1}) = \frac{dA}{dt}A(t)^{-1} + A(t)\frac{d}{dt}A(t)^{-1}$$

Thus:

$$\frac{d}{dt}A(t)^{-1} = -A(t)^{-1}\frac{dA}{dt}A(t)^{-1}$$

Next we consider the exponential operator, where $\exp : \mathrm{Mat}_{\mathbb{R}}(n \times n) \to \mathrm{Mat}_{\mathbb{R}}(n \times n)$, with $A \to \sum_{n=0}^{\infty} \frac{1}{n!}A^n$, we have $\exp(A + B) = \exp(A)\exp(B)$ if $AB = BA$, so that:

$$\frac{d}{dt}\exp(tA) = A\exp(tA) = \exp(tA)A,$$

and that $D\exp(\mathbf{0})(B) = \frac{d}{dt}_{t=0}\exp(tB) = B$, so that $D\exp(\mathbf{0})$ is the identity map on $\mathrm{Mat}_{\mathbb{R}}(n \times n)$. More generally:

$$D\exp(A)(B) = \frac{d}{dt}_{t=0}\exp(A + tB)$$

Since $\exp(-A)\exp(A + tB) - I = \int_0^1 \frac{d}{ds}(\exp(-sA)\exp(s(A + tB)))ds = \int_0^1$

15 Day 15 (10.16) — Approximation in n-Variables

Theorem 15.1: L

et $U \subseteq \mathbb{R}$, and $f \in C^N(U)$, then, for $c \in U$:

1. There is an unique polynomial $p(x)$ of degree $\leq N$ such that:

$$f^{(j)}(c) = p^{(j)}(c),$$

for $j \in \{0, \cdots, N\}$.
It's given by the formula:

$$p(c) = \sum_{j=0}^{N} \frac{f^{(j)}(c)}{j!}(x-c)^j.$$

2. The remainder term $r(x) = f(x) - p(x)$ satisfies $\lim_{x \to c} \frac{r(x)}{(x-c)^N} = 0$.

Example 15.2: S

uppose $f'(c) = 0$, then:

1. If $f''(c) > 0$, then f has local main.
2. If $f''(c) = 0$, then f could be either/neither.

Consider the smallest $N > 0$ such that $f^{(N)}(c) \neq 0$, then:

$$f(x) = f(c) + \frac{f^{(N)}(c)}{N!}(x-c)^N + r(x).$$

Hence, we have:

$$\frac{f(x) - f(c)}{(x-c)^N} = \frac{f^N(c)}{N!} + \frac{r(x)}{(x-c)^N}$$

And if N is even, then this expression obtains its local minima or maxima depending on the sign of $f^{(N)}(c)$, and if N is odd, then $f(x) - f(c)$ changes sign. The only case that this doesn't work is when all $f^{(j)} = 0$.

Definition 15.3: Polynomial of degree N

Let $x_1, \cdots, x_n \in \mathbb{R}^n$, then a function:

$$p(x_1, \cdots, x_n) = \sum a_{\alpha_1, \cdots, \alpha_n} x_1^{\alpha_1}, \cdots, x_n^{\alpha_n}$$

is a polynomial of degree N, and the sum is $(\alpha_1, \cdots, \alpha_n)$ where $\alpha_i \in \mathbb{Z}_{\geq 0}$ and $\alpha_1 + \cdots + \alpha_N \leq N$. And the coefficients are recovered using partial derivatives, where:

$$\frac{\partial \beta_1}{\partial x_1^{\beta_1}} \cdots \frac{\partial \beta_n}{\partial x_n^{\beta_n}}(x_1^{\alpha_1} \cdots x_n^{\alpha_n}) = \begin{cases} 0, \text{ if } (\beta_1, \cdots, \beta_n) \neq (\alpha_1, \cdots, \alpha_n) \\ \alpha_1! \alpha_2! \cdots \alpha_n!, \text{ if } (\beta_1, \cdots, \beta_n) = (\alpha_1, \cdots, \alpha_n) \end{cases}$$

Definition 15.4: Multi-Index Notation

$x = (x_1, \cdots, x_n)$, where $x_i \in \mathbb{R}$.
$\alpha = (\alpha_1, \cdots, \alpha_n)$, $\alpha_i \in \mathbb{Z}_{\geq \not<}$.
$|\alpha| = \alpha_1 + \cdots + \alpha_n$
$x^\alpha = x_1^{\alpha_1}, ..., x_n^{\alpha_n}$
$\frac{\partial |\alpha|}{\partial x^\alpha} = \frac{\partial \alpha_1}{\partial x_1^{\alpha_1}}, \cdots, \frac{\partial \alpha_n}{\partial x_n^{\alpha_n}}$
$f^{(\alpha)} = \frac{\partial |\alpha|}{\partial x^\alpha} f$.

So, we write:

$$p(x) = \sum_{|\alpha| \leq N} a_\alpha x^\alpha, \ \frac{\partial |\beta|}{\partial x^\beta} x^\alpha = \begin{cases} 0, \text{ if } \beta \neq \alpha \\ \alpha!, \text{ if } \beta = \alpha \end{cases}$$

And the coefficient of $p(x)$ is:

$$a_\alpha = \frac{1}{\alpha!} \frac{\partial |\alpha|}{\partial x^\alpha} p(x)$$

Theorem 15.5: Taylor Theorem

Let $U \subseteq \mathbb{R}^n, f \in C^N(U)$, then for $c \in U$:

1. There is an unique polynomial $p(x)$ of degree $\leq N$ such that:

$$f^{(\alpha)}(c) = p^{(\alpha)}(c), \ |\alpha| \leq N.$$

   It's given by the formula:

$$p(x) = \sum_{|\alpha| \leq N} \frac{f^{(\alpha)}(c)}{\alpha!} (x - c)^\alpha.$$

2. The remainder $r(x) = f(x) - p(x)$ satisfies $\lim_{x \to c} \frac{r(x)}{\|x - c\|^N} = 0$.

Then, we'll discurss the critical points of the function $f : U \to \mathbb{R}$, i.e., whem $\nabla f(c) = 0$, we consider the higher order derivatives, e.g. $\frac{\partial^2 f}{\partial x_i \partial y_j}(c)$.

16 Day 16 (10.19) — Taylor Expansion and Hessian

Lemma 16.1. Supposer that $r \in C^N(U)$, with $r^{(\alpha)}(c) = 0$ for all $|\alpha| \leq N$. Then:

$$\lim_{x \to c} \frac{r(x)}{\|x - c\|^N} = 0.$$

Proof. We have to show:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall y \in U, \|y - c\| > \delta \Rightarrow |r(x)| \leq \varepsilon \|x - c\|^N.$$

We want to use the Mean Value Theorem. Given $x \in U$, consider:

$$\phi(t) = r((1-t)c + tx, \ t \in [0,1].$$

By the Mean Value Theore, there exists $t' \in [0,1]$ such that $\phi(1) - \phi(0) = \phi'(t')$.
Put $y' = (1-t')c + t'x$, then:

$$\phi(1) = r(x), \phi(0) = r(c), \phi(t') = r(y').$$

Therefore:

$$\begin{aligned} r(x) = \phi(1) - \phi(0) &= \phi'(t') \\ &= \sum_{j=1}^{n} \frac{\partial r}{\partial x_j} y'(x_j - c_j) \end{aligned}$$

Hence:

$$\begin{aligned} |r(x)| &\leq \sum_{j=1}^{n} \left| \frac{\partial r}{\partial x_j}(y') \right| |x_j - c_j| \\ &\leq \|x - c\| \sum_{j=1}^{n} \left| \frac{\partial r}{\partial x_j}(y') \right|. \end{aligned}$$

Let $\varepsilon > 0$ be given, the by induction, we assume that the statement is true for $N - 1$. For $N \geq 1$, we apply to $\dfrac{\partial r}{\partial x_j} \in C^{N-1}(U)$. Hence, there exists $\delta > 0$ such that:

$$\|y - c\| < \delta \Rightarrow \left| \frac{\partial r}{\partial x_j}(y) \right| \leq \frac{\varepsilon}{n} \|y - c\|^{N-1}.$$

Then :

$$|r(y)| \leq \|y - c\| \sum_{j=1}^{n} \frac{\varepsilon}{n} \|y - c\|^{N-1} = \varepsilon \|y - c\|^N.$$

□

Example 16.2

WTS the Taylor Polynomial of degree 4 of the function $f(x) = \dfrac{1}{1+x^2+y^3}$.

Since $\frac{1}{1-t} = \sum_{i=0}^{\infty} t^i$, we have:

$$\begin{aligned} f(x) &= \frac{1}{1+x^2+y^3} = \frac{1}{1-(-x^2-y^3)} \\ &= \sum_{i=0}^{\infty} (-x^2-y^3)^i \\ &= 1 - x^2 - y^3 + x^4 + \dots \end{aligned}$$

Example 16.3

WTS the Taylor Polynomial of the function $f(x,y) = \frac{1}{x^2+y^3}$ at $c = (1,0)$. We write $x = 1+u$, then:

$$\begin{aligned} f(x,y) &= \frac{1}{(1+u)^2+y^2} = \frac{1}{1+(u^2+2u+y^2)} \\ &= \frac{1}{1-(-u^2-2u-y^2)} \\ &= \sum_{i=0}^{\infty} (-u^2-2u-y^2)^i \\ &= \sum_{i=0}^{\infty} [-(x-1)^2 - 2(x-1) - y^2]^i \\ &= 1 - 2(x-1) + 3(x-1)^2 - 4(x-1)^3 - y^3 \end{aligned}$$

Then we apply the Taylor Theorem to the critical point behavior of a function, where we know that if $f \in C^1(U)$ has the local extrema at $c \in U$, then $\nabla f(c) = 0$. If $f \in C^2(U)$, then:

$$f(x) = f(c) + \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}(c)(x_j - c_j) + \frac{1}{2}\sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i}(x_i - c_i) + r(x).$$

The for $n = 2$, we have:

$$f(x,y) = f(r.s) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(r,s)(x-r)^2 + \frac{\partial^2 f}{\partial x \partial y}(c)(x-r)(y-s) + \frac{1}{2}\frac{\partial^2 f}{\partial y^2}(r,s)(y-s)^2 + \dots$$

General case:

$$f(x) = f(c) + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(c)(x_i - c_i) + \frac{1}{2}\sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i \partial y_j}(c)(x_i - c_i)(x_j - c_j),$$

Definition 16.4: Hessian

Suppose $U \subseteq \mathbb{R}^n$, $c \in U$, and $\nabla f(c) = 0$, then the Hessian of f at c is the matrix:

$$H_y = \frac{\partial^2 f}{\partial x_i \partial y_j}(c).$$

In this case:

$$f(x) = f(c) + \frac{1}{2}\sum_{i,j=1}^{n} H_y(x_i - c_i)(x_j - c_j) + r(x).$$

Using the inner product notation:

$$f(x) = f(c) + \frac{1}{2}\langle H(x-c), (x-c)\rangle + r(x).$$

Thus, we obtain a lemma:

> **Lemma 16.5.** If H is positive definite, then $\langle H(x-c), (x-c)\rangle > 0$ for $x - c = 0$, which means that f attains its local minima at c. Similarly, if H is negative definite, then f attains its local maxima at c, and otherwise f is at its saddle point at c.

17 Day 17 (10.21) — Critical Point Analysis

Proposition 17.1 (Necessary Conditions For Local Extrema)**.** Let $U \subseteq \mathbb{R}^n$ be open, and let $f \in C^1(U)$. If f has a local maxima or local minima at $c \in U$, then:

$$\frac{\partial f}{\partial x_j}(c) = 0,\ j \in \{1, \cdots, n\}.$$

Equivalently, $Df(c) = 0$, i,e, $\nabla f(c) = 0$.

Proof. If f has a local extrema at c, then each of the functions:

$$t \mapsto f(c + te_j),$$

has a local extrema at $t = 0$, then the t-derivative of this function must vanish at $t = 0$, and this means:

$$\frac{\partial f}{\partial x_j}(c) = 0.$$

□

Proposition 17.2. Let $A \in \mathrm{Mat}_{\mathbb{R}}(n \times n)$ be a real symmetric matrix of size n, then:

1. There is an orthonormal basis $v_1, \cdots, v_n \in \mathbb{R}^n$ consisting of eigenvectors of A, thus:

   $$Av_i = \lambda_i v_i,$$

   where $\lambda_i \in \mathbb{R}$ are the eigenvalues, and:

   $$\langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1, i = j \\ 0, i \neq j \end{cases}.$$

   In particular, A has n real eigenvalues λ_i.

2. Let S be the orthogonal matrix having $v_1, \cdots, v_n$ as its columns, we have that:

   $$A = SDS^{\top},$$

   where D is the diagonal matrix.

3. Let $q_A(x) = \langle Ax, x \rangle$ be the quadratic form of defined by A on x, then use the orthonormal bases $v_1, \cdots, v_n$ to write:

   $$x = \sum_{i=1}^{n} y_i v_i,$$

   and we write $x = Sy$, then $Ax = \sum_{i=1}^{n} \lambda_i y_i v_i$, then:

   $$q_A(x) = \sum_{i=1}^{n} \lambda_i y_i^2$$

Theorem 17.3

The maxima (resp. minima) of the restriction of q_A to the unit sphere:

$$\{x : \|x\| = 1\},$$

is the largest eigenvalue (resp. smallest eigenvalue) of A. The corresponding x is a normalized eigenvector for that eigenvalue.

Proof. Since S is an orthogonal matrix, we have:

$$\|x\|^2 = \|Sy\|^2 = \|y\|^2 = \sum_i y_i^2.$$

We write $t_i = y_i^2 \geq 0$, then $\|y\| = 1$ means $\sum t_i = 1$. Hence, the maxima of

$$\sum_{i=1}^{n} \lambda_i t_i,$$

is obtained by taking $t_i = 1$ for the largest eigenvalue and $t_i = 0$ for all other eigenvalues. Conversely, if $\langle Ax, x\rangle$ equals to the largest eigenvalue, then x must be a linear combination of eigenvectors for that eigenvalue, for otherwise the sum

$$\sum_{i=1}^{n} y_i^2 \lambda_i,$$

would be smaller than the largest eigenvalue. The minima proof is similar. □

Definition 17.4: Positive Definite Matrix

A matrix $A \in \mathrm{Mat}_{\mathbb{R}}(n \times n)$ is called positive definite if the quadratic forms for all $x \neq 0$ satisfy:

$$q_A(x) > 0.$$

Definition 17.5: Negative Definite Matrix

A matrix $A \in \mathrm{Mat}_{\mathbb{R}}(n \times n)$ is called negative definite if the quadratic forms for all $x \neq 0$ satisfy:

$$q_A(x) < 0.$$

Corollary 17.6. A symmetric matrix A is positive definite if all of its eigenvalues are greater than 0, and negative definite if all of its eigenvalues are smaller than 0.

Proposition 17.7 (Level Sets)**.** For $n = 2$, we obtain the level sets:

$$q_A^{-1}(\mu) = \{x : q_A(x) = \mu\}.$$

If both eigenvalues are positive, the level sets for $\mu > 0$ are ellipses, the level sets for $\mu < 0$ are empty and $q_A^{-1}(0) = 0$. The conditions for the negative eigenvalues are reversed.

And if there are both positive and negative eigenvalues, the non-zero level sets are hyperbolas, while $q_A^{-1}(0)$ consists of two intersecting lines.
And if the eigenvalues are all zero, then the quadratic form does not depend on the corresponding coordinate.

Proposition 17.8 (Sufficient Conditions For Local Extrema)**.** Let $f \in C^2(U)$, and $c \in U$. If $\nabla f(c) = 0$, and $Hf(c)$ is positive definite (resp. negative definite), then f has a local minima (resp. maxima) at c.

18 Day 18 (10.23) — Critical Points and Level Sets

Proposition 18.1. Let $U \subseteq \mathbb{R}^n$ be an open set, and let $f : U \to \mathbb{R}$ be a function. Let $c \in U$. Then:

$$f(x) = f(c) + \nabla f(c)(x-c) + \frac{1}{2}Hf(c)(x-c)^2 + r(x).$$

And the necessary condition for a critical point is $\nabla f(c) = 0$, in this condition, we have:

$$f(x) \approx f(c) + \frac{1}{2}Hf(c)(x-c)^2,$$

where $Hf(c) = \frac{\partial^2 f}{\partial x_j \partial x_j}(c)$. If $Hf(c)$ is positive definite, then f attains its local minima at c, and the level sets are ellipses, thus:

$$\frac{1}{2}yHf(c)(y) = \frac{1}{2}(\lambda_1 y_1^2 + \lambda_2 y_2^2),$$

with eigenvalues $\lambda_1, \lambda_2 > 0$.

There, let the two eigenvectors of $Hf(c)$ be v_1, v_2 respectively, then $\lambda_1 > \lambda_2 > 0$ means f's level sets has longer axis on the direction of eigenvector v_1.

Let $W_+ \subseteq \mathbb{R}^n$ be the sum of eigenspace for positive eigenvalues of $Hf(c)$, and W_- be the sum of eigenspace for negative eigenvalues of $Hf(c)$, then $\mathbb{R}^n = W_+ \oplus W_-$.

Then $q_H(y) = \langle Hy, y\rangle$ is positive definite on W_+ and negative definite on W_-. So, restriction to W_+ has local minima, and restriction to W_- has local maxima.

Example 18.2

Let $f(x,y) = x^3 + y^3 - 3xy$, then:

$$\nabla f(x,y) = \begin{pmatrix} 3x^2 - 3y \\ 3y^2 - 3x \end{pmatrix},\ Hf(x,y) = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}.$$

Therefore, let $\nabla f(c) = 0$, then $c = (0,0)$ or $(1,1)$.
At $c = (0,0)$,

$$Hf(x,y) = \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix},$$

and $\lambda_1 = 3, v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \lambda_2 = -3, v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
Hence, $(0,0)$ is a saddle point, with hyperbolic level sets, and $f(x,y) \approx -3xy$.
At $c = (1,1)$,

$$Hf(x,y) = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix},$$

and $\lambda_1 = 9, v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \lambda_2 = 3, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
Hence, $(0,0)$ is a local minima, with elliptic level sets, and $f(x,y) \approx 2 - 3xy$.

19 Day 19 (11.2) — Inverse Function Theorem

Inverse function theorem is about "coordinate changes", for example: the polar coordinate

$$x = r\cos\theta, y = r\sin\theta$$

can be represented by the map $f : \mathbb{R}^2 \to \mathbb{R}^2$, $(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$.
On $\mathbb{R}^2\backslash\{(0,0)\}$, we can locally invert this map, eg. solve for r, θ ($r = \sqrt{x^2+y^2}$), but what about the more general coordinate changes?

We consider $(u_1 \cdots, u_n) = \phi(x_1, \cdots, x_n)$, where $\phi : U \to \mathbb{R}^n$ and $c \in U$. We want to solve for $(x_1, \cdots, x_n)$ to make $(x_1, \cdots, x_n) = \psi(u_1, \cdots, u_n)$, thus we want to find an open neighborhood V of $\phi(c)$ and $\psi : V \to \mathbb{R}^n$ such that:

$$\phi \circ \psi = \mathrm{id}_V .$$

By Chain Rule, if $\phi \in C^1(U, \mathbb{R}^n)$, we expect $\psi \in C^1(V, \mathbb{R}^n)$ to get:

$$D_c\phi D_{\phi(c)}\psi = I.$$

Thus, the necessary condition is $D\phi(c)$ being invertible.

Theorem 19.1: Inverse Function Theorem

Suppose $f \in C^k(U, \mathbb{R}^n)$ with $k \geq 1$. Let $c \in U$ and $Df(c)$ be invertible, then there exists open neighborhoods U_1, V_1 of c and $f(c)$ such that:

$$c \in U_1 \subseteq U,\ f(c) \in V_1 \subseteq \mathbb{R}^n,$$

and there exists an inverse function of f:

$$g \in C^k(V, \mathbb{R}^n),$$

such that $f(U_1) = V_1,\ g(V_1) = U_1$, and $f \circ g = \mathrm{id}_{V_1},\ g \circ f = \mathrm{id}_{U_1}$.

Lemma 19.2. Suppose $h \in C^1(U, \mathbb{R}^m)$ with $h(0) = 0$ and $Dh(0) = 0$, then for every $\delta > 0$, there exists $r > 0$ such that for all $x, y \in \overline{B_r(0)}$, we have:

$$\|h(x) - h(y)\| \leq \delta\|x - y\|.$$

Proof. We WTS that there exists $r > 0$ such that for all $x \in \overline{B_r(0)}$,

$$\|Dh(x)v\| \leq \delta\|v\|$$

for all $v \in \mathbb{R}^n$, where $v = \sum_{i=1}^n v_i e_i$, with standard basis $e_1, \cdots, e_n$. Then:

$$\|Dh(x)v\| = \left\|\sum_{i=1}^n v_i[Dh(x)e_i]\right\| \leq \sum_{i=1}^n |v_i|\|Dh(x)e_i\| \leq \|v\|\sum_{i=1}^n \|Dh(x)e_i\|.$$

But $x \mapsto \|Dh(x)e_i\|$ is continuous, vanishes at 0, hence there exists $r > 0$ such that $\|x\| \leq r \Rightarrow \|Dh(x)e_i\| < \dfrac{\delta}{n}$. Thus:

$$\|Dh(x)v\| \leq \delta\|v\|.$$

□

Proof of Inverse Function Theorem. We assume that $c = 0$ and $f(0) = 0$, then:

$$f(x) = f(0) + Df(0) + r(x) = Df(0) + r(x).$$

We let $Df(0) + A$, and replace f with $\tilde{f} = f \circ A^{-1}$, $y \mapsto f(A^{-1}y)$, then:

$$D\tilde{f}(0) = Df(0) \circ A^{-1} = I.$$

Using this, we may also assume that $Df(0) = I$, then by the Lemma, $f(x) = x + h(x)$.

Step 1 : **Local Injectivity** – WTS there exists $r > 0$ such that for all $x, y \in \overline{B_r(0)}$, $x \neq y \Rightarrow f(x) \neq f(y)$. We use the Lemma with $\delta = \dfrac{1}{2}$, and take $r > 0$, then:

$$\begin{aligned} \|f(x) - f(y)\| &= \|x - y + h(x) - h(y)\| \\ &= \|x - y\| - \|h(x) - h(y)\| \\ &= \frac{1}{2}\|x - y\|. \end{aligned}$$

And in particular, $x \neq y \Rightarrow f(x) \neq f(y)$.
We take $y = x + tv$, and let $t \to 0$, then:

$$\frac{\|f(x) - f(x + tv)\|}{\|tv\|} \geq \frac{1}{2} \Rightarrow \frac{\|Df(x)v\|}{\|v\|} \geq \frac{1}{2}.$$

Hence, we obtain:

$$\|Df(x)v\| \geq \frac{1}{2}\|v\|, \ \forall v \in \mathbb{R}^n.$$

Hence, $Df(x)$ is invertible for all $x \in B_r(0)$.

Step 2 : **Local Surjectivity** – WTS $f(B_r(0)) \supseteq B_{\frac{r}{4}}(0)$, we let $y_0 \in B_{\frac{r}{4}}(0)$ and consider the function $\varphi : \overline{B_r(0)} \to \mathbb{R}$, where $\varphi(x) = \|f(x) - y_0\|^2$. Since $\overline{B_r(0)}$ is compact, by Extreme Value Theorem, $\varphi(x)$ takes a minima. Thus for $\|x\| = r$, we have:

$$\|f(x) - y_0\| \geq \|f(x)\| - \|y_0\| \geq \frac{r}{2} - \|y_0\| > \frac{r}{4} > \|y_0\|,$$

which gives that the minima of φ is not on the boundary, so the minima is in the interior, so we use the criteria for local minima, we have that if φ takes on a local minima at x_0, then:

$$\begin{aligned} 0 = D\varphi(x_0)v &= \left.\frac{d}{dt}\right|_{t=0} \varphi(x_0 + tv) \\ &= \left.\frac{d}{dt}\right|_{t=0} \langle f(x_0 + tv) - y_0, f(x_0 + tv) - y_0\rangle \\ &= 2\langle Df(x_0)v, f(x_0) - y_0\rangle \end{aligned}$$

Since $Df(x_0)$ is invertible, we have that $f(x_0) = y_0$. □

20 Day 20 (11.4) — Inverse Function Theorem, Diffeomorphism and Morse Lemma

Proof of Inverse Function Theorem (continued). We continue to proof the theorem.

Step 3 : Differentiability of the Inverse – Let $V_1 = B_{\frac{r}{4}}(0)$ and $U_1 = f^{-1}(V_1) \cap B_r(0)$. Since f is a bijection on the restricted domain, we let $g : V_1 \to U_1$ be the inverse, then we have:

$$\|g(w) - g(z)\| \leq 2\|w - z\|,$$

in particular, g is continuous. Then we want to show that g is differentiable at the point $y_0 = f(x_0) \in V_1$. By definition of the derivative of f:

$$f(x) = f(x_0) + Df(x_0)(x - x_0) + r(x), \ \lim_{x \to x_0} \frac{r(x)}{\|x - x_0\|} = 0.$$

In terms of $y = f(x)$ and $y_0 = f(x_0)$, this says:

$$y = y_0 + Df(x_0)(g(y) - g(y_0)) + r(g(y)),$$

which we can solve for $g(y)$ using invertibility of $Df(x_0)$:

$$g(y) = g(y_0) + (Df(x_0))^{-1}(y - y_0) - (Df(x_0))^{-1} r(g(y)).$$

Then we have:

$$\frac{\|r(g(y))\|}{\|y - y_0\|} \leq 2 \frac{\|r(g(y))\|}{\|g(y) - g(y_0)\|},$$

and by continuity of g:

$$\lim_{y \to y_0} \frac{\|r(g(y))\|}{\|g(y) - g(y_0)\|} = \lim_{x \to x_0} \frac{\|r(x)\|}{\|x - x_0\|} = 0,$$

so also $\lim\limits_{y \to y_0} \dfrac{\|r(g(y))\|}{\|y - y_0\|} = 0$. Hence, g is differentiable at y_0, which gives:

$$Dg(y_0) = (Df(g(y_0)))^{-1}.$$

Step 4 : Regularity – – Suppose that f is of class C^k, WTS g is of class C^k. On V_1 we have:

$$Dg = (Df \circ g)^{-1},$$

which means that:

$$Dg = \text{Inv} \circ Df \circ g,$$

where $g : V_1 \to U_1 \subseteq \mathbb{R}^n$, and $Df : U_1 \to \text{Mat}_{\mathbb{R}}(n \times n)^{\times}$, where we denote $\text{Mat}_{\mathbb{R}}(n \times n)^{\times}$ the open subset of matrices A with $det(A) \neq 0$, and matrix inversion:

$$\text{Inv} : \text{Mat}_{\mathbb{R}}(n \times n)^{\times} \to \text{Mat}_{\mathbb{R}}(n \times n)^{\times}, \ A \mapsto A^{-1}.$$

The map Inv is of class C^{∞}, and Df is of class C^{k-1}, and f is of class C^k. Therefore, g is of class C^j for any $j \leq k - 1$, then Dg is of class C^j, so g is of class C^{j+1}. Since g is continuous, we then obtain inductively that g is of class C^k. Thus we finish the proof of Inverse Function Theorem. □

Definition 20.1: C^k-Diffeomorphism

Let $U \subseteq \mathbb{R}^n$ be open, and $f : U \to \mathbb{R}^n$ be a function of class C^k. Then f is called a C^k-diffeomorphism onto an open subset $V \subseteq \mathbb{R}^n$ if it admits a C^k inverse $g : V \to \mathbb{R}^n$, that:

$$f \circ g = \mathrm{id}_V, \ g \circ f = \mathrm{id}_U .$$

Definition 20.2: Local C^k-Diffeomorphism

Let $U \subseteq \mathbb{R}^n$ be open, and $f : U \to \mathbb{R}^n$ be a function of class C^k. Then f is called a local C^k-diffeomorphism if for all $c \in U$, there are open neighborhoods

$$U_1 \subseteq U \ni c, \ V_1 \subseteq \mathbb{R}^n \ni f(c),$$

such that $f|_{U_1} : U_1 \to V_1$ is a C^k-diffeomorphism.

Corollary 20.3. A map $f : U \to \mathbb{R}^n$ is a local diffeomorphism if and only if the Jacobian matrix $Df(x)$ is invertible for all $x \in U$.

Example 20.4: Polar Coordinate

Consider the map:

$$\phi : \{(r, \theta) \mid r > 0\} \to \mathbb{R}^2, \ (r, \theta) \mapsto (r \cos\theta, r \sin\theta),$$

then:

$$D\phi(r, \theta) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \ \det D\phi(r, \theta) = r > 0.$$

Hence, ϕ is a local diffeomorphism.

Lemma 20.5 (Morse Lemma)**.** Let $f \in C^\infty(U)$ be a smooth function with a critical point at $c \in U$, thus $\nabla f(c) = 0$. Suppose also that the Hessian $H(c)$ is nondegenerate, with r positive eigenvalues and $n - r$ negative eigenvalues, then there exists a C^∞ coordinate change $\phi : U_1 \to V_1$ on an open neighborhood U_1 of c, with $\phi(c) = 0$, such that the resulting function $\tilde{f} = f \circ \phi^{-1}$ is of the form:

$$\tilde{f}(y) = f(c) + y_1^2 + \cdots + y_r^2 - y_{r+1}^2 - \cdots - y_n^2.$$

21 Day 21 (11.6) — Implicit Function Theorem

22 Day 22 (11.9) — Rank Theorem

Notes & Questions

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